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HOME SITE SCENT MARKING WITH URINE AND AN ORAL SECRETION IN THE GOLDEN SPINY MOUSE (*ACOMYS RUSSATUS*)

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ABSTRACT

Captive golden spiny mice (*Acomys russatus*) use odors from different sources to mark their home area. The marking behavior is sexually dimorphic, with males marking more frequently than most females. This difference may be related to mobility; females being more sedentary than males. In both sexes, the entrance to the nesting place was the main site of marking. The animals repeatedly marked this area by smearing the entrance with a thick oral secretion and by depositing small urine marks. Under the harsh ecological conditions under which *A. russatus* lives, nesting sites are presumably subject to fierce competition. The role of this chemical signaling in the temporal segregation between the two sympatric species, *A. russatus* and *A. cahirinus*, is discussed. Golden spiny mice also mark their food and leftovers with oral scent. This marking may have a social function in making food detection easier for the kin.

INTRODUCTION

In the extremely hot and arid rocky regions of the Rift Valley and Negev Desert in Israel, two spiny mice of the genus *Acomys*, *A. russatus* (the golden spiny mouse) and *A. cahirinus* (the common spiny mouse), share the same habitat. However, these species are temporally segregated. While *A. cahirinus* is nocturnal, *A. russatus* is diurnal, despite the hostile environment (Shkolnik and Borut, 1969).

Experiments in the field (Shkolnik, 1971) and in the laboratory (Rubal et al., 1992) demonstrated that when *A. cahirinus* is removed from the common habitat, *A. russatus* becomes nocturnal. *A. russatus*, therefore, appears to be excluded from nocturnal activity by its competitor.

It has been shown that when a pair of spiny mice, composed of one of each species, is kept in an enclosure with two artificial burrows, each animal occupies a separate nest. They are relatively tolerant of each other, except at the nesting places. At these sites, aggression is initiated mainly by *A. cahirinus*. Usually, nocturnal *A. russatus* loses its nest to *A. cahirinus*, and is forced to rest outside during the day. Diurnal *A. russatus*, on the

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other hand, may reoccupy its original nest at night (Haim and Rozenfeld, 1994). Why *A. cahirinus* selects a nesting place already occupied by *A. russatus*, rather than a free one, remains unknown. However, these results strongly suggest that in the field, a suitable nesting site that provides good insulation against extreme temperatures may be a major source of competition between these sympatric species.

The diurnal activity rhythm of *A. russatus* appears to be induced by chemical signals from the other species. The urine and feces of *A. cahirinus* induce a shift in the daily rhythms of oxygen consumption (Haim and Fluxman, 1992) and times of activity (Haim and Rozenfeld, 1993).

Preliminary tests indicate that the urine and feces of *A. cahirinus* do not immediately induce a marking response in *A. russatus*. Conspecific odors, however, trigger an overmarking behavior in *A. russatus*, probably with a ventral gland (Rozenfeld and Haim, 1992). These observations indicate that *A. russatus* emits species-specific pheromones, and suggest that an allomone carried by the urine and feces of *A. cahirinus* has a primer effect on *A. russatus*. This may be a unique situation since most of the known priming chemical signals are species-specific and modulate reproductive physiology (Vandenbergh, 1988; Muller-Schwarze, 1989). This allomone may maintain temporal isolation between the two species so that they can occupy the same shelters at different times. Such an isolating mechanism could reduce the high costs involved in the physical defence of a resting place in the hostile desert environment.

The observation that, in the laboratory, *A. cahirinus* selected nests of *A. russatus* during daytime clearly indicates that it is able to discriminate between its own original nesting place and an *A. russatus* nest. Since both nesting places were physically identical, we expect that olfactory cues from *A. russatus* play an important role in this discrimination. The nesting area, therefore, should be the main place of scent marking by this species.

The purpose of the present investigation is to assess the importance and modalities of scent marking in *A. russatus* of both sexes.

In the first experiment, we determined whether isolated spiny mice of both sexes scent mark at the same rate and at the same sites, and whether the frequency of scent marking decreases with familiarization with their enclosure.

A second experiment was conducted to determine if females of the golden spiny mouse signal their reproductive readiness by scent marking, as do those of several other rodent species (Johnston, 1983). Since before the present work, nothing was known about the length of the ovarian cycle in *A. russatus*, we investigated the frequency of marking by females in relation to aspects of their vaginal smears for a period of 30 days.

In the course of these experiments, it was observed that spiny mice encountering sunflower seeds pick them up, first with their mouths, then with their front legs, and turn the seeds over in their mouths before eating them. Some seeds, however, are abandoned rather than eaten. While exploring their enclosure, males often stop at these previously manipulated seeds to sniff them briefly. This olfactory investigation suggested that the handled seeds could carry some specific odors. To test this hypothesis, the response of a male to a sunflower seed that had been previously handled by a foreign conspecific was analyzed in a third experiment.

MATERIALS AND METHODS

ANIMALS

Sixteen adult golden spiny mice (9 males and 7 females) were obtained from a laboratory breeding colony in Israel (The University of Haifa at Oranim). The colony was established from individuals caught in the Dead Sea area.

The animals were individually housed in polycarbonate cages ($28 \times 26 \times 16$ cm) until the beginning of the experiments. They were provided with a mixture of commercial hamster chow and sunflower seeds, fresh carrots, and wood shavings. They were kept at 27 °C in a reversed light cycle (12L:12D); during the "bright" phase, the enclosures were illuminated with fluorescent tubes and during the "dark" phase, by photographic dark-room orange bulbs.

OBSERVATION ENCLOSURES

The enclosures were large modular observation pens ($57 \times 57 \times 48$ cm) made of three aluminum walls and a front glass window through which behavior could be observed and video-recorded. The pen was connected to a Y-shaped artificial plaster burrow with three short galleries converging on a central nest chamber. The paired branches of the Y were closed with wire netting, and the unpaired branch was connected to the observation pen through a glass tube (Rasmont and Rozenfeld, 1987). Although golden spiny mice do not dig burrows in the field, they readily accepted this structure for their rudimentary nest. For simplicity, we refer to the whole plaster device as the "nest," and to the entrance of the glass tube as the nest "entrance."

The aluminum floor of the pen was lined with blank newspaper. Before each observation, the flooring paper was replaced with filter paper (Schleicher and Schull) to collect the urine marks.

The pen was provided with an open glass vial (7 cm high and 7 cm wide) serving as a food trough and a running wheel. These items were placed against two walls at equal distances from the entrance of the nest. The pen was covered with a wire netting.

EXPERIMENTAL PROCEDURES

On day 0 of the experiments, each animal was transferred from its small cage to the observation enclosure where it remained until the end of the experiments, 14 days for males, and 40 days for females.

EXPERIMENT 1

On day 0, the animals were isolated, each in a single enclosure. Their behavior was observed the following day (day 1) and once again on day 8. After each observation, the urine marks deposited during the previous 24 h were collected for counting.

For each animal, the observations started at the same time of the day, 1–2 h before the beginning of the dark phase.

Recordings: Behavior outside the nesting place was video-recorded for 30 min. The recorded locomotory and scent-marking movements were quantified with a computer.

The following events were analyzed.

- (1) Locomotory patterns: the amount of time each animal spent at three main sites: (a) near the nest, i.e., with any part of its body within 3 cm of the nest entrance, (b) near the food trough, and (c) on the running wheel. The frequencies of its departures from the nest area were also determined.
- (2) Scent-marking behavior: The numbers and duration of the (a) urine marking and (b) oral marking episodes were recorded.

Counting urine marks and oral marking: For each animal, the frequencies of urine marking were determined by counting the marks deposited on the flooring filter paper in four quadrates (N: nest, F: food trough, W: running wheel, and E: empty), during a 30-min observation on days 1 and 8, and also during the 24 h following these observations. The urine marks were viewed and counted under UV light (Rozenfeld et al., 1987).

The frequencies of oral marking were determined directly from the video records made during the observation sessions. They occur only on sites N, F, and W; in the empty quadrate, there was nothing to nibble on.

EXPERIMENT 2

This experiment involved only 6 of the 7 females used in Experiment 1. For 30 days after the end of Experiment 1, i.e., from day 9 till day 39, vaginal smears of the females were examined daily and flooring filter papers were collected 24 h later for counting the urine marks. Vaginal smears were obtained by flushing the vagina with physiological saline. The cells were stained with hemalin eosin and examined microscopically for the presence of specific cell types. Three stages were identified: proestrus, estrus, and metestrus. Proestrus was diagnosed from a smear containing 60–80% nucleated epithelial cells and 20–40% leukocytes, estrus from a smear containing exclusively cornified epithelial cells, and metestrus by the presence of leukocytes only.

EXPERIMENT 3

After experiment 1, i.e., from day 8, the 9 males were maintained in the familiar environment without removing their flooring paper. On days 12 and 14, we recorded their response to the presence of the following stimuli: (1) a sunflower seed that had been freshly manipulated but not eaten by a strange male, and (2) a seed manipulated by the test animal itself.

The tests were carried out in the following way: 19 seeds that had never been in contact with a spiny mouse were scattered in a 10-cm circle drawn next to the trough. The stimulus seed was collected with a clean forceps and deposited among these seeds. The order of presentation of the stimuli and controls was random.

STATISTICAL ANALYSIS

As our samples are small and the probability distribution of our measurements is unknown, we represented all our results by the median, and the dispersion by the extremes (Neave and Worthington, 1988). Where necessary, we assessed statistical significance by using the randomization test (Siegel, 1956), in the Statmode Two program (Coleman and Coleman, 1987).

RESULTS

LOCOMOTORY ACTIVITY

The sexes behaved differently from each other. Males rapidly and repeatedly patrolled the whole enclosure. They briefly stopped to sniff the floor and the various objects, and to deposit scent marks. Males always ate in the immediate vicinity of the trough. With the exception of the trough, they never stayed outside of their nest for long. Females, in contrast, remained at the nest entrance for long periods of time, engaging in grooming only. When moving from one place to another in the enclosure, they usually return first to the nest entrance. Occasionally, they carry seeds to their nest. On the whole, females are much more sedentary than males.

Experimental results are summarized in Table 1. Though the total time spent out of the nest was not significantly different between the sexes (column T), males spent significantly less time near the nest entrance (column N) and left it more frequently (column D) than females. They also spent significantly more time at the food trough (column F) and less at the wheel (column W). In general, males spent the greatest part of their activity time near the food trough (column F), while females spent it near the nest (column N).

SCENT-MARKING ACTIVITY

Urine marking: Males and females pass large quantities of urine in the corners of their pen. Only a small number of urine drops are deposited near their nest. While moving around and sniffing at the substrate, males drag the tip of their prepuce on the floor, leaving long, fine urine trails (Fig. 1). Females, in contrast, stop during marking, flattening their belly on the floor and pointing their clitoris backwards to spray a few very small urine drops (Fig. 2). Males also deposit urine traces on protruding objects. As these behavioral patterns are clearly different from excretion, and between the sexes, they may be considered as scent marking.

Oral marking: Animals of both sexes were often seen nibbling at the rim of the nest entrance, then rapidly moving the muzzle sideways along the edge of the glass tube (Fig. 3). Males, but not females, also nibbled on different objects protruding in the

Table 1
Locomotory patterns during a 30-min observation period

	T	N		F		W		D
	durations	durations	N/T	durations	F/T	durations	W/T	numbers
♂	1,193.1	246.5(**)	0.21	544.3(**)	0.46	164.1(*)	0.14	54(**)
(n = 9)	<i>1,013–1,800.0</i>	<i>158.2–382.1</i>		<i>357.4–741.6</i>		<i>0–267</i>		<i>49–178</i>
♀	1,299.2	460.8(**)	0.35	383.1(**)	0.29	276(*)	0.21	25(**)
(n = 7)	<i>1,100.2–1,394.0</i>	<i>385.8–559.1</i>		<i>236–449.2</i>		<i>159.4–341</i>		<i>14–58</i>

Total duration (in seconds) spent out of the nest (T); time spent in the nesting area (N), near the food trough (F), on the running wheel (W), and ratios of these durations to T; numbers of departures from the nest area (D). The medians are given in regular type, the extreme values in italics. Within columns, * = $p < 0.05$; ** = $p < 0.005$ (two-tailed randomization test).



Fig. 1. Male dragging his penis during urine marking.



Fig. 2. Female pointing her clitoris backwards during urine marking.

enclosure. To study this behavior in detail, the pens of two males were provided with a clean fine glass tube, fastened on the front glass window. The behavior at this tube was video-recorded and analyzed frame by frame. The spiny mice were seen nibbling at the tip of the tube, then pressing a corner of their mouth all along the tube so as to spread oral secretion. At the end of the experiments (on day 14), the rim of the nest entrance of the males was covered with a thick layer of pinkish secretion (Fig. 4).

Urine and oral marking were generally preceded and followed by sniffing. The reverse was not true; sniffing at a scent mark was not necessarily followed by marking.

The results of the observations on urine and oral marking are summarized in Table 2. In both sexes, urine marking and oral marking were significantly ($p < 0.005$) more frequent at the nest entrance (row N) than near the food trough (row F) or at the wheel

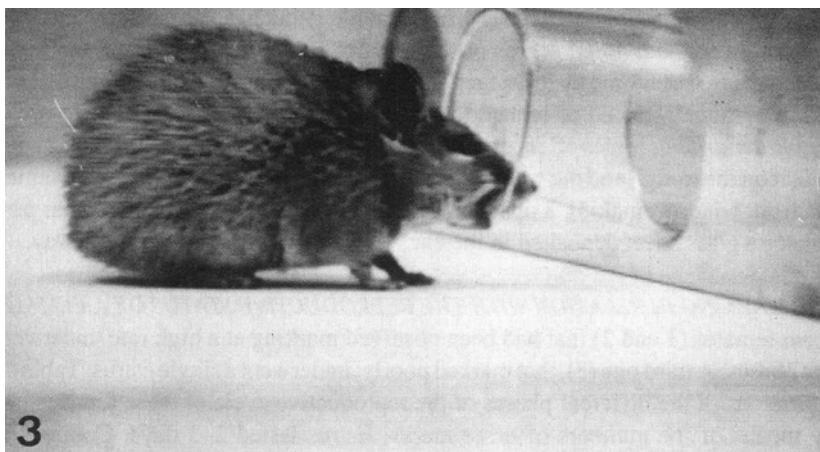


Fig. 3. Oral marking at the nest entrance.

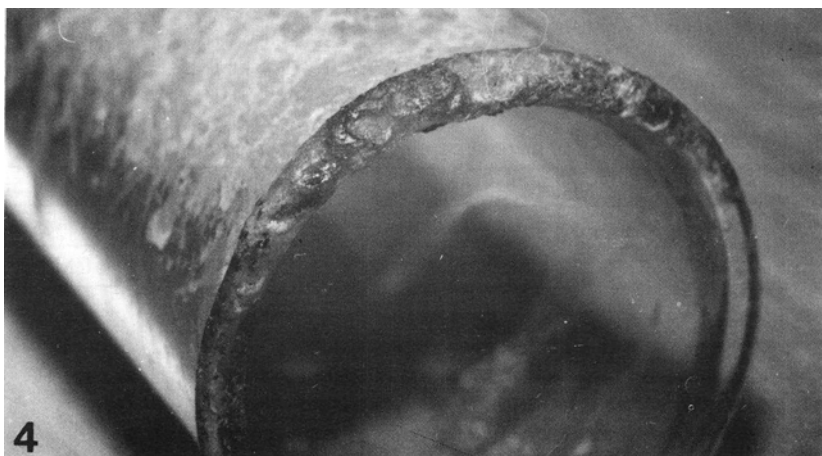


Fig. 4. Oral secretion deposited on the rim of the nest entrance.

(row W). Overall, males did more urine and oral markings than females (row T). There was no significant difference ($p > 0.05$) in either sex between the markings on days 1 and 8.

As the frequencies of urine markings were very low during the 30 min observations, we collected the urine marks left during the 24 h following days 1 and 8. The data are presented in Table 3. There was no significant decrease in the total numbers of marks deposited by both males and females between days 1 and 8.

In both sexes, quadrat N is more marked than the others. Next, in descending order, come the food quadrat (F), the wheel quadrat (W), and the empty quadrat (E). These results are the same as those in Table 2. The proportion of marks deposited in quadrat N is greater for females than for males, which is consistent with the fact that females spend more time near the nest entrance.

Row T of Table 3 shows that overall, males marked much more than females. The large dispersion of the individual scores of the females is due to two of them marking very profusely, more than some males in fact. This observation suggests that a high rate of urine marking by females could be related to a specific stage of their reproductive cycle.

Plantar marking: Males often jumped around in the observation pen, kicking with their hind feet on the trough and the walls. Three males repeated this activity on the same place of the front window, making it opaque with an accumulation of whitish plantar prints. A pebble or a glass sheet deposited in the enclosure were treated in the same way.

URINE MARKING IN RELATION WITH THE REPRODUCTIVE STATUS OF A FEMALE

Two females (1 and 2) that had been observed marking at a high rate underwent two estrus phases. A third one (3), that marked poorly, underwent a single estrus. Table 4 gives the durations of the different phases of the reproductive cycle of these females, and the daily means of the numbers of urine marks. Estrus lasted 2–3 days. Contrary to our prediction, no consistent change in the daily urine marking was observed in the course of the different stages of the reproductive cycle. Three other females, who were poor markers, remained in metestrus during the entire period.

Table 2
Urine and oral markings during 30-min observation periods

	Day 1				Day 8			
	σ (n = 9)		φ (n = 7)		σ (n = 9)		φ (n = 7)	
	UM	OM	UM	OM	UM	OM	UM	OM
N	19 _(a,b) <i>4–36</i>	35 _(c,d) <i>20–51</i>	4 _(e,f) <i>0–8</i>	12 _(g,h) <i>5–15</i>	27 <i>7–42</i>	42 <i>16–59</i>	6 <i>0–14</i>	13 <i>5–18</i>
F	12 _(a) <i>4–25</i>	13 _(c) <i>5–39</i>	1 _(e) <i>0–3</i>	0 _(g) <i>0–2</i>	6 <i>2–19</i>	9 <i>0–15</i>	0 <i>0–2</i>	0 <i>0</i>
W	8 _(b) <i>0–16</i>	13 _(d) <i>0–40</i>	0 _(f) <i>0–1</i>	0 _(h) <i>0</i>	6 <i>2–12</i>	5 <i>1–18</i>	0 <i>0–9</i>	0 <i>0</i>
E	1 <i>0–9</i>	–	0 <i>0</i>	–	2 <i>0–11</i>	–	0 <i>0–1</i>	–
T	39 <i>14–67</i>	42 <i>31–90</i>	5 <i>1–12</i>	16 <i>5–23</i>	39 <i>19–66</i>	53 <i>33–85</i>	9 <i>0–23</i>	13 <i>5–18</i>

Urine marks (UM) and oral markings (OM) near the nest entrance (N), the food trough (F), the running wheel (W), and in the empty quadrat (E), by 9 males and 7 females, during 30-min observations on days 1 and 8. Row T gives the totals or urine marks in the four quadrates, and of oral markings in the three quadrates containing items. The medians are given in regular type, the extreme values in italics. Within columns, results marked with the same lower-case index are significantly different, with $p < 0.05$, those marked with the same capital index are so with $p < 0.005$ (two-tailed randomization test).

Table 3
Urine markings during 24-h observation periods

	Day 1				Day 8			
	σ ($n = 9$)		φ ($n = 7$)		σ ($n = 9$)		φ ($n = 7$)	
	n	n/ Σ	n	n/ Σ	n	n/ Σ	n	n/ Σ
N	273 <i>57-482</i>	0.39	34 <i>2-368</i>	0.60	292 <i>63-391</i>	0.45	26 <i>2-292</i>	0.72
F	221 <i>24-349</i>	0.31	15 <i>0-130</i>	0.26	201 <i>32-266</i>	0.31	5 <i>2-170</i>	0.14
W	161 <i>12-227</i>	0.23	6 <i>0-126</i>	0.11	107 <i>14-198</i>	0.17	3 <i>1-39</i>	0.08
E	54 <i>7-195</i>	0.08	2 <i>0-124</i>	0.04	47 <i>7-93</i>	0.07	2 <i>0-18</i>	0.06
Σ	709		57		647		36	
T	738 <i>100-1112</i>		50 <i>6-503</i>		707 <i>120-852</i>		35 <i>5-504</i>	

Median numbers (n) of urine marks deposited in the quadrates nest (N), food trough (F), running wheel (W), empty (E), and in the four quadrates together (T), by 9 males and 7 females, during 24-h periods starting on days 1 and 8. Row Σ gives the sums of the medians in each quadrate (which are different from the medians of the total markings T). Columns n/ Σ give the ratios of the median numbers of markings in each quadrate to Σ . The medians are given in regular type, the extreme values in italics.

RESPONSE OF MALES TO SEEDS PREVIOUSLY HANDLED BY ANOTHER MALE

To test the hypothesis that food handling is associated with scent marking, each of the nine males was presented with 20 seeds, one of which had been previously handled either by the subject itself or by a different male. The results show that the animals spent significantly ($p < 0.005$) less time sniffing seeds they had handled previously (2.0 s, range: 0-9.2) than seeds handled by a strange male (16.4 s, range: 6.6-28.1). After the initial sniffing, the animal takes the "strange" seed into its mouth and repeatedly licks and sniffs it before leaving it on the floor. The median number of licking bouts was 6 (3-13), and the median time spent licking was 18.1 s (2.8-29.1). The response of the spiny mice to their "own" seed was very low, the median number of licking bouts being 0 (0-2) and the median time was 0 s (0-5.6). Both differences are statistically significant ($p < 0.05$). It may therefore be assumed that the spiny mouse is able to recognize a seed that has been previously handled by a strange conspecific.

DISCUSSION

Our experiments demonstrate that solitary golden spiny mice profusely scent-marked their environment. By jumping onto or kicking against various surfaces or items, males

Table 4

Durations (d) of the phases of the reproductive cycle of females 1 to 3, and daily mean numbers (u) of urine marks deposited during these phases

	♀ 1		♀ 2		♀ 3	
	d	u	d	u	d	u
Metestrus	11	858.6 (92.3)	15	353.6 (63.6)	10	21.3 (7.6)
Proestrus	2	414.5 (109.6)	1	385	3	21 (7.8)
Estrus	2	357.5 (30.4)	3	393.5 (71.4)	2	15.5 (0.7)

Durations are in days; for numbers of urine marks, means are in regular type, S.D. in italics.

deposit a whitish secretion from the plantar surface of their hind feet. Sweat glands have been found in the foot pads of other rodent species including desert rodents (Quay, 1965a), but their possible role in communication remains unclear. It is, however, possible that footprints are olfactory tracks helping in the animal's orientation.

Males and females marked their environment with a visible thick oral secretion and with urine. Males marked more often than most females, dispersing marks over almost any physical feature in their home area, while females marked almost exclusively at the entrance of their nest. This difference in the spatial distribution of scent marking may be explained by different movement patterns, females being more sedentary than males. The same difference occurred when golden spiny mice were kept in couples. Although they shared the same nesting box, the female left it less often than the male. It may be expected that in the field, males have larger home ranges or territories than females, and these are marked with dispersed scent posts.

With both sexes, the entrance of the nest emerged as a focus of repeated marking, mainly with oral secretion but also with urine. As suggested by Eisenberg and Kleiman (1973), these olfactory marks probably serve to establish and maintain the animal's own "odor field", and to remind the subject of its own place. They could also be used to warn conspecific neighbors that the place is occupied (Ralls, 1971; Johnston, 1983; Vandenberg, 1988). These hypothetical functions require that an animal may at least be able to distinguish between the odor of its own marks and those of aliens. Preliminary tests indicate that the golden spiny is able to do so (Rozenfeld and Haim, 1992).

The sources of odors left at the nest entrance are still unknown. The oral secretion, which is mixed with saliva, probably comes from a gland that is visible inside the mouth corner. In *Acomys dimidiatus*, large mucous glands were found in the oral angle by Quay (1965b). To some extent, the oral marking was similar to the "mouth-cheek rubbing" described in different species of squirrels (Halpin, 1985). Preputial and clitoral glands are a possible source of chemical signals carried by the urine in rodents (Brown, 1985), but we have not succeeded in finding any glands under dissection.

The three females that underwent estrus did not change their rate of urine marking throughout the entire cycle. This observation, however, should be extended to more females, especially breeding ones, before concluding that urine marking is independent of ovarian hormones. In female Mongolian gerbils (*Meriones unguiculatus*), scent

marking associated with a ventral gland also varies among individual females, increasing from mid-pregnancy till early lactation, and decreasing to lower levels at weaning (Wallace et al., 1973). In this species, scent marking clearly plays an important role in mother-infant interactions.

The golden spiny mice scent mark food and leftovers. When introduced into the home area of a strange male, these marked objects were rapidly detected and licked. This response indicates that an individual oral odor had been deposited during previous handling. Licking may serve at least 2 functions: (1) to transport chemical cues to vomeronasal receptors, and (2) to mask the original odor. As the male golden spiny mouse eats its food on the spot, without gathering it, seed marking may play an important social role in the detection and selection of food by other members of his group. In *Acomys cahirinus*, McFadyen-Ketchum and Porter (1989) demonstrated the transmission of food preferences from mothers to their offspring by oral cues.

In conclusion, *A. russatus* deposits odors from different sources, presumably as a means of self-identification and of warning conspecific neighbors. However, at least some of these signals could also be meaningful to other species, for instance *A. cahirinus*, which may subsequently use them to select a suitable resting site during daytime.

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